

MA2221: Foundational Mathematics for ML

Assignment (19–01–2026)

Instructor: **Biswarup Biswas****Instructions:**

- This assignment was conducted during the **lab session**.
- Students were required to type all answers using **Markdown with LaTeX**.
- Correct mathematical typesetting was the primary objective.
- Mathematical understanding was **not required** unless explicitly stated.

1. Inline Mathematics

Typeset the following sentences exactly as written, ensuring that inline mathematics renders correctly.

- (a) The quadratic polynomial $ax^2 + bx + c$ has discriminant $b^2 - 4ac$.
- (b) The solution of $ax^2 + bx + c = 0$ is given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- (c) For small h , the finite difference approximation is

$$\frac{f(x+h) - f(x)}{h}.$$

2. Displayed Equations

Typeset the following equations using displayed mathematics.

- (a) Taylor expansion of the exponential function:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

- (b) A definite integral:

$$\int_0^1 x^2 dx = \frac{1}{3}.$$

3. Aligned Equations

Use the **align** environment to typeset the following multi-line expressions.

- (a)

$$\begin{aligned} (x+1)^2 &= x^2 + 2x + 1 & (1) \\ &= x^2 + 2x + 1 & (2) \end{aligned}$$

- (b)

$$\begin{aligned} \int_0^1 (3x^2 + 2x + 1) dx &= [x^3 + x^2 + x]_0^1 & (3) \\ &= 3 & (4) \end{aligned}$$

4. Mathematical Paragraph

You are **not required to understand** the mathematics below. Your task is to typeset the paragraph correctly.

Newton's method is an iterative algorithm for solving nonlinear equations. Starting from an initial guess x_0 , the method generates a sequence $\{x_n\}$ using the formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Under suitable conditions, the sequence converges to a root of the equation $f(x) = 0$.

5. Common Mathematical Symbols

Typeset the following expressions correctly.

(a) Limit:

$$\lim_{x \rightarrow 0^+} \frac{\log x}{\log(1/x)} = -1$$

(b) Summation:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

(c) Partial derivative:

$$\frac{\partial u}{\partial x}$$

End of Assignment